Electric Vehicles Aggregator Optimization – A Fast and Solver-Free Solution Method

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1 Mixed–Integer Models Considered and Proposed Solution Algorithm

2 Electric Vehicles Charging Coordination

Outline

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Large Scale Optimization – Problem Structure Considered

- we consider the problem

$$\mathcal{P} : \begin{cases} \min & \sum_i c_i x_i \\ \text{s.t.} & \sum_i H_i x_i \leq b \\ & x_i \in X_i \qquad\qquad i = 1, \ldots, \mathrm{I} \end{cases}$$

  where

$$X_i = \{x_i \in \mathbb{R}^{r_i} \times \mathbb{Z}^{z_i} \mid A_i x_i \leq d_i\}$$

- large collection of subsystems
  - subsystem model is X_i (mixed–integer)
  - coupled by shared resources $\rightarrow$ coupling constraints $\sum_i H_i x_i \leq b$
- # of subsystems $|\mathrm{I}| \gg$ # of coupling constraints m

Problem's Decomposition

Obtain decomposition using duality:

$$\begin{array}{ll} \min_x & \sum_{i\in \mathrm{I}} c_i x_i \\ \text{s.t.} & \sum_{i\in \mathrm{I}} H_i x_i \leq b \\ & x_i \in X_i \end{array} \quad \Rightarrow \quad \begin{array}{ll} \min_x & \sum_{i\in \mathrm{I}} c_i x_i + \lambda'(\sum_{i\in \mathrm{I}} H_i x_i - b) \\ \text{s.t.} & x_i \in X_i \end{array}$$

$$\Rightarrow \underbrace{\sum_{i\in \mathrm{I}} \min_{x_i \in X_i} \left\{ c_i x_i + \lambda'(H_i x_i) \right\} - \lambda' b}_{\doteq d(\lambda)}$$

- Lagrangian dual (or outer) problem:

$$\mathcal{D} : \left\{ \begin{array}{ll} \max & d(\lambda) \\ \text{s.t.} & \lambda \geq 0 \end{array} \right.$$

Solutions to the inner problem

$$d(\lambda) = \sum_{i \in \mathrm{I}} \min_{x_i \in X_i} \left\{ c_i x_i + \lambda'(H_i x_i) \right\} - \lambda' b$$

Consider

$$x_i(\lambda^\star) \in \arg\min_{x_i \in X_i} \left\{ c_i x_i + \lambda^\star (H_i x_i) \right\}$$

as candidate solution to $\mathcal{P}$

- generally **infeasible** in the MIP case !
  - violate coupling constraints

Primal Recovery Scheme

- we show that in $x(\lambda^\star)$ at most m subproblems may be "problematic"
  - Shapley–Folkman Theorem (Shapley won Nobel Prize in Eco)
  - related to **bound on duality gap** [Ekeland '76, Bertsekas '83]
- so we propose to consider

$$\overline{\mathcal{P}} : \begin{cases} \min_x & \sum_{i \in \mathrm{I}} c_i x_i \\ \text{s.t.} & \sum_{i \in \mathrm{I}} H_i x_i \leq b - \rho \\ & x_i \in X_i \qquad \forall i \in \mathrm{I}, \end{cases}$$

where

$$\rho = m \cdot \max_{i \in \mathrm{I}} \left(\max_{x_i \in X_i} H_i x_i - \min_{x_i \in X_i} H_i x_i \right)$$

Theorem [Vujanic '14]

Then $x(\overline{\lambda}^\star)$ is feasible for $\mathcal{P}$. [under some uniqueness assumption]

Performance of the Recovered Solutions

- under some technical assumption, ...

Theorem [Vujanic '14]

The recovered solution $x(\bar{\lambda}^\star)$ is feasible and satisfies

$$J_{\mathcal{P}}(x(\bar{\lambda}^\star)) - J^\star_{\mathcal{P}} \leq (m + ||\rho||_\infty/\zeta) \cdot \max_{i \in \mathrm{I}} \left(\max_{x_i \in X_i} c_i x_i - \min_{x_i \in X_i} c_i x_i \right)$$

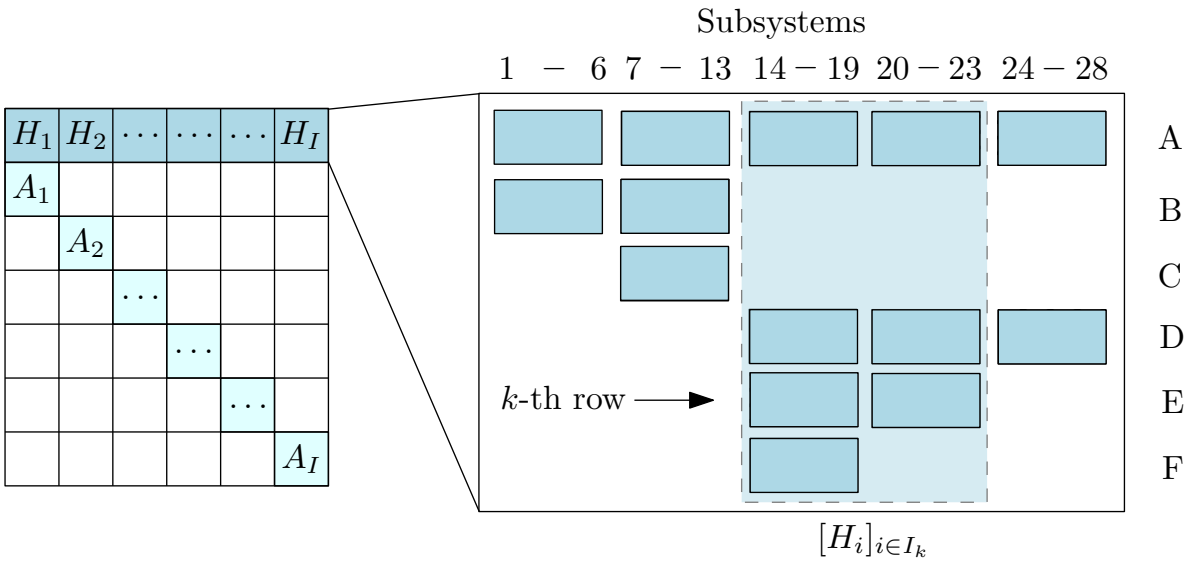
- if $J^\star_{\mathcal{P}}$ grows linearly with $|\mathrm{I}|$, m is fixed and X_i uniformly bounded

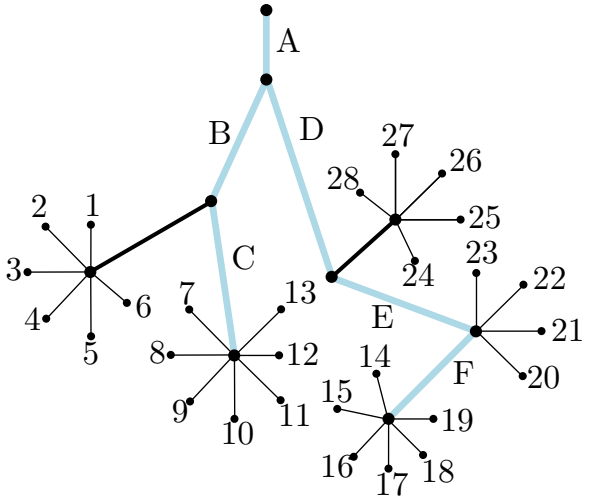
$$\frac{J(x(\bar{\lambda}^\star)) - J^\star_{\mathcal{P}}}{J^\star_{\mathcal{P}}} \to 0 \quad \text{as} \quad |\mathrm{I}| \to \infty$$

Improving Approximations – Conservatism Reduction

ρ scales with m but **not** with I – want to keep it as small as possible

- when couplings are determined by certain network topologies





- can safely use $\mathrm{rank}([H_i]_{i \in \mathrm{I}_k})$ instead of m
- generally possible to use $\mathrm{rank}(H)$ instead of m

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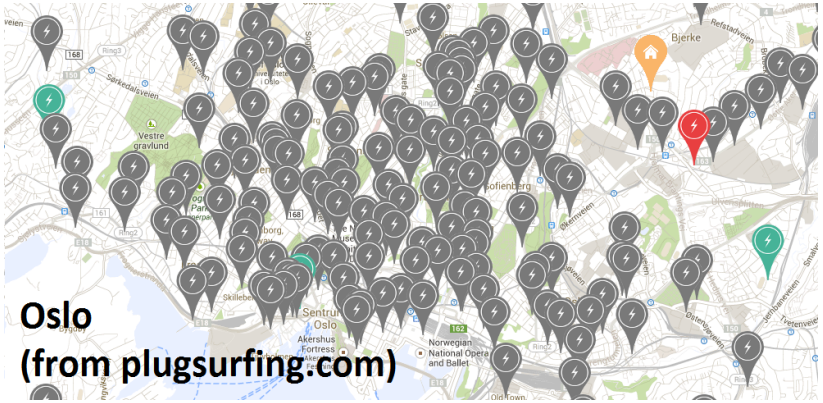
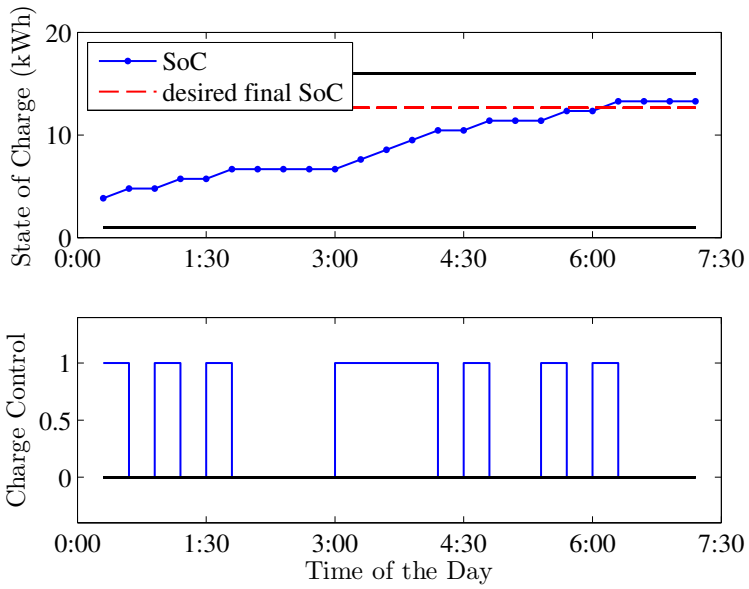
Electric Vehicle (EV) Charging Coordination

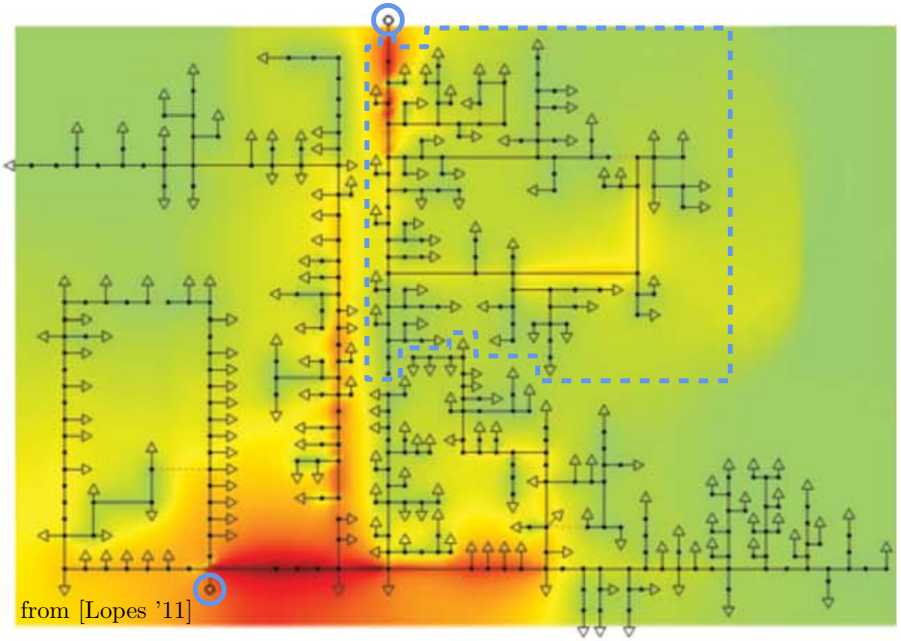
- expected increase of EV presence
- substantial additional stress on network & equipment
  $\Rightarrow$ need charging coordination
- network administrator (DSO) can't manage each unit individually $\Rightarrow$ **EV aggregator**



Aggregator's Role

- aggregator's **control task** is to assign to each EV the time slots when charging can occur
- compatibly with...
- **local requirements**
  - required final State of Charge
  - fixed charge rates
  - battery capacity limits
- **global objectives**
  - network congestion avoidance (limits set by DSO)
  - "valley fill", cost min., ...





Control Strategy

- cast as large optimization problem

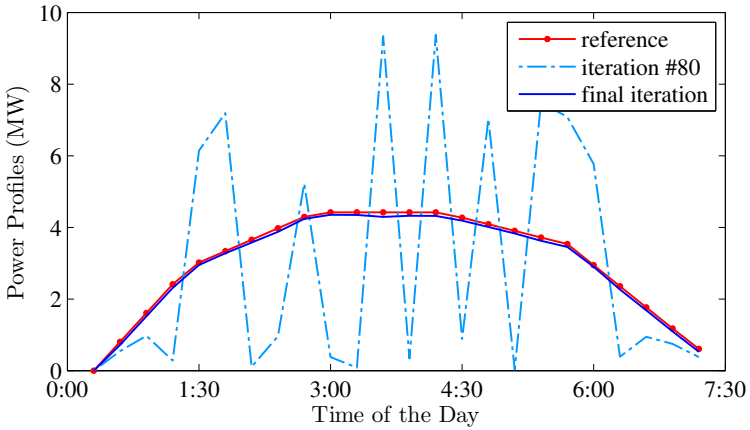
$$\begin{cases} \underset{u_i}{\text{minimize}} & \sum_{k=0}^{N-1} \left| \sum_{i \in I} P_i u_i[k] - P^{\text{ref}}[k] \right| \\ \text{subject to} & \sum_{i \in I^{\text{cr}}} P_i u_i \leq P^{\text{max}} \\ & e_i[0] = E_i^{\text{init}} \\ & e_i[k+1] = e_i[k] + (P_i \Delta T \zeta_i) u_i[k] \\ & e_i[N_i] \geq E_i^{\text{ref}} \\ & e_i[k] \leq E_i^{\text{max}} \\ & u_i \in \{0,1\}^N, \end{cases} \leftrightarrow \mathcal{P} : \begin{cases} \min & \sum_i c_i x_i \\ \text{s.t.} & \sum_i H_i x_i \leq b \\ & x_i \in X_i \end{cases}$$

- with
  - $u_i[k] \in \{0,1\}$: charge decision at step k for EV $i \in \mathrm{I}$
  - $e_i[k]$: state of charge of battery $i \in \mathrm{I}$
  - P_i: charge rate
  - $E_i^{\text{init}}, E_i^{\text{ref}}, E_i^{\text{max}}$: initial, final required and maxium state of charge
  - ζ_i: conversion losses

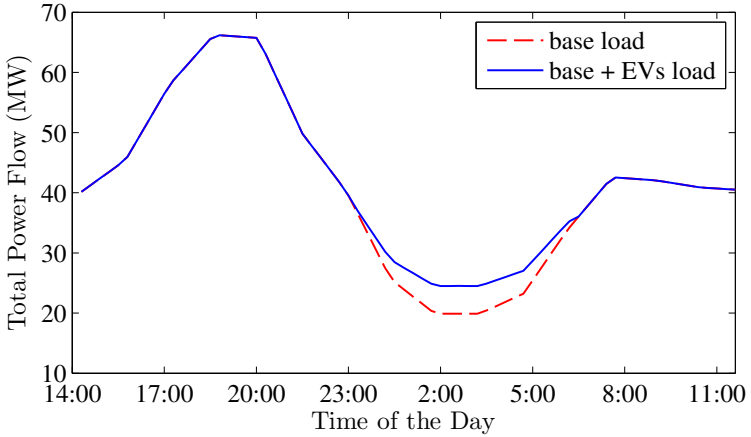
Computational Experiments

- solve using proposed method: duality+contraction
  - support extensions (e.g., vehicle–to–grid "V2G")
- population up to 10'000 EVs
- computation times $\leq$ 10 sec (charge only)
  - greedy subproblem structure
  - no external solver needed

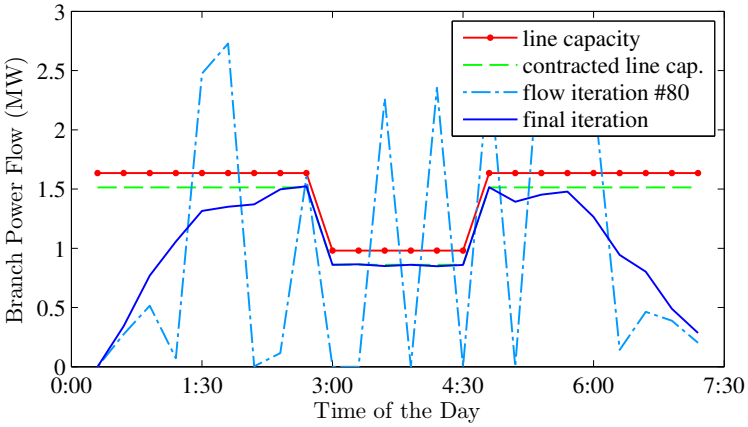
Solutions – Charge and V2G



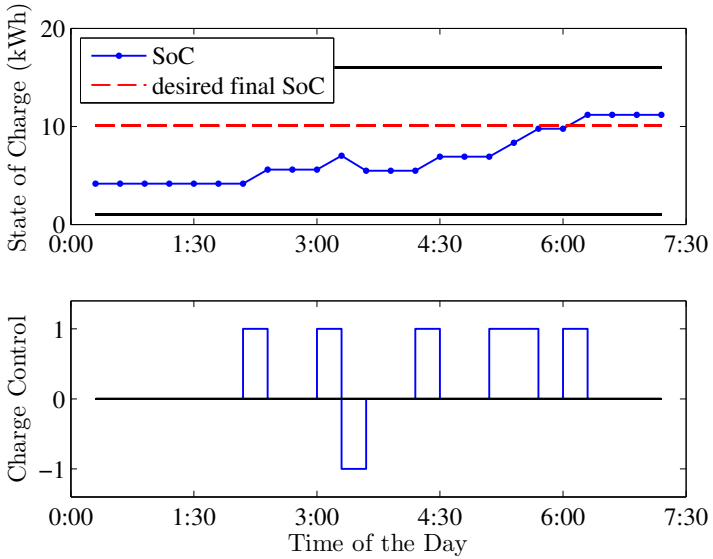
(a) reference tracking



(b) resulting "valley fill"



(c) network limits



(d) local requirements

Other Example Applications

- supply chains optimization – partial shipments [Vujanic '14b]
- power systems operation
  - control of TCLs
  - large fleet of generators
- portfolio optimization for small investors
- ...

Questions?

All technical details/proofs on **arXiv**
`http://arxiv.org/abs/1411.1973`

- [Ekeland '76] J. P. Aubin and I. Ekeland, *Estimates of the duality gap in nonconvex optimization*, Mathematics of Operations Research **1** (1976), no. 3, 225-–245.
- [Bertsekas '83] Dimitri P. Bertsekas, G. Lauer, N. Sandell, and T. Posbergh, *Optimal short-term scheduling of large-scale power systems*, IEEE Transactions on Automatic Control **28** (1983), no. 1, 1– 11.
- [Dawande '06] Milind Dawande, Srinagesh Gavirneni, and Sridhar Tayur, *Effective heuristics for multiproduct partial shipment models*, Operations Research **54** (2006), no. 2, 337–352 (en).
- [Vujanic '14] R. Vujanic, P. Mohajerin Esfahani, P. Goulart, S. Mariethoz and M. Morari, *A Decomposition Method for Large Scale MILPs, with Performance Guarantees and a Power System Application*, submitted to Automatica (2014).
- [Vujanic '14b] R. Vujanic, P. Goulart, M. Morari, *Large Scale Mixed-Integer Optimization: a Solution Method with Supply Chain Applications*, Mediterranean Conference on Control & Automation, 2014
- [Lopes '11] J. Lopes, F. Soares, and P. Almeida, *Integration of Electric Vehicles in the Electric Power System*, Proceedings of the IEEE, 2011, p.168-183.